JDCBL: A Joint Dynamic Contexts and Balanced Learning Framework for Real-World Fabric Defect Detection

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***Abstract*— Fabric defect detection is crucially essential for maintaining fabric quality. Existing intelligent detection systems, however, often fall short in real scenarios due to significant variations in defect sizes, which typically manifest as small and elongated structures. In addition, insufficient samples in real scenarios result in a long-tail distribution in training data, posing challenges to the balanced data prerequisites of having collecting balanced data for deep learning methods. This article introduces the joint dynamic contexts and balanced learning (JDCBL) framework to cope with these challenges. Within this framework, contextual transformer block (CoTBlock) is designed to capture both global and fine-grained features of fabric defects simultaneously, which forms contextual transformer network (CoTNet). This modification boosts the interactivity of contextual information between input keys and effectively enhances self-attention learning. Additionally, class average and class complement strategies balanced contrastive loss (BCL) amplify the exposure of tail-class defects so they can be accurately identified. Extensive validation on the Tianchi fabric dataset demonstrates that the JDCBL framework surpasses other leading methods in multiple defect detection tasks, achieving a MAP50 of 39.3%—over 5% higher than competing methods.**

***Index Terms*— Attention mechanism, balanced learning, defect detection, fabric, fine-grained, long-tailed learning.**

I. Introduction

FABRIC defect detection is a crucial task of quality control in the textile industry. Fabric production is complex, and various defect types can occur, leading to profit losses ranging from 35% to 55% [1], [2]. Therefore, implementing efficient automatic detection and accurate identification of these defects is very importance for improving textile enterprises' product quality and capacity.

In previous studies, researchers have focused on classical methods such as statistical approaches [3], structural methods [4], spectral methods [5], and model-based techniques [6] to facilitate the intelligent detection of defects. These methods use the explicit design or selection of operators to extract features, and they have the advantageous of being high-speed, lightweight, and easy to interpret [7]. For instance, Raheja et al. [8] leveraged the gray level co-occurrence matrix (GLCM) to extract texture features from fabric images and developed a real-time defect detection system based on an embedded DSP platform. Liu et al. [9] introduced a dual sparse low-rank decomposition method for defect detection in complex, irregularly printed fabrics. Sadaghiyanfam [10] proposed an automated implementation of a system based on GLCM, known as FDDS. However, these methods are highly dependent on prior knowledge, are technically demanding, and involve lengthy development cycles.

In recent years, deep learning-based methods have developed rapidly. Classifiers are trained with large amounts of labeled data to learn feature representations corresponding to different defects and establish a mapping relation between the feature and the labeled spaces, thereby completing defect detection and classification tasks. Networks such as VGG [11] ResNet [12] and others have promoted deep learning methods in research and applications in many fields. For example, Jin et al. [13] used a feature pyramid network (FPN) for defect localization and successfully implemented surface defect detection on rails. Shang et al. [14] proposed a method for detecting damage to aircraft engine blades based on Mask R-CNN. Zeng et al. [15] proposed an improved ABFPN based on FPN, which was successfully applied to defect detection on PCB boards. Deep learning methods can realize an end-to-end detection process that achieves relatively high recognition accuracy and adapts to detections in various complex scenarios.

In real scenarios, fabric defects present various features resulting from different causes, leading to significant variations in detection efficiency and accuracy during online inspection. This article analyzes the features of fabric defects in real scenarios, specifically the area ratio and aspect ratio. As shown in Fig. 1, where Ar = defect area/img area represents the area ratio, and Lr = min{length/width, width/length} represents the aspect ratio. Defects such as Hole and Stain, with relatively large area ratios and aspect ratios, are easy to detect. In contrast, defects like "Hundred feet" and "Drop warp" display characteristics of small area ratios and extreme aspect ratios. For example, the area ratio of a knot defect is only 0.0003, while the extreme aspect ratio of a Drop warp defect reaches 0.05. This makes such defects difficult to detect, highlighting the challenge posed by extreme data in real-world fabric scenarios.

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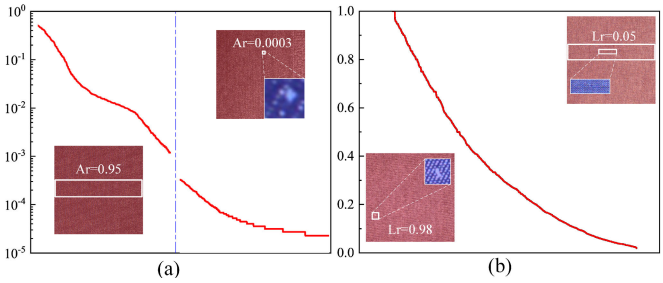


Fig. 1. Distribution of fabric defect features in real scenes is illustrated as follows: (a) area ratio of the defect relative to the entire image. It highlights two extreme cases: one defect with an area ratio of 0.95 and another with 0.0003. For clarity, only a portion of the large-area defect is displayed, while a magnified view is provided for the small-area defect and (b) aspect ratio of the defect, featuring a schematic of two cases: one with an aspect ratio of 0.98 and another with 0.05. Both defect features are magnified for better visualization.

Due to the interference of complex background textures, defects in extreme scenarios show significant intra-class variations and minimal inter-class differences, leading to slight differences in the features of various defect types. In recent years, researchers have focused on richer fine-grained features by constructing channel [16], [17] and spatial attention [18], [19], [20]. Moreover, the diverse causes of defects result in significant variations in the data volume of different types of defects, creating a long-tail distribution in training datasets. Take the defects in weft yarn as an example: these defects, though less frequent found in real scenarios and often appearing in the tail-class data, can have serious consequences. Once not promptly detected and dealt with by immediate machine shutdowns, these defects could lead to severe production accidents.

Therefore, this article proposes the joint dynamic contexts and balanced learning (JDCBL) framework. This framework focuses on rich contextual information to generates attention matrices and build a practical mapping relation of features, allowing the feature extraction network to fully represent the fine-grained features of defects with low area ratio and high aspect ratio. Additionally, the framework enhances the detection capabilities for tail classes without compromising the accuracy of head classes in fabric defect detection. The contributions of this article are summarized as follows.

1) This article introduces the JDCBL framework. It has been successfully applied to fabric defect detection tasks. It achieved SOTA performance in detection tasks for the tail classes and the overall dataset. The framework achieved 39.3% in mAP50, more than 5% higher than the comparative methods.
2) To address the detection of fine and elongated defects, we proposed the use of contextual transformer networks (CoTNets) as the feature extractor. This structure, by integrating self-attention mechanisms, focuses on fine-grained features and rich contextual relationships, thus extracting spatially relevant refined features.
3) To address the imbalance in fabric data, we introduced a new balanced loss function designed to equalize the contribution of different class data during gradient updates. This function uses class average and complement strategies to increase the exposure of tail classes, thereby effectively facilitating the learning of these lower frequency classes. Moreover, training with balanced contrastive loss (BCL) significantly enhances the model's ability to distinguish between different classes.

The remainder of this article is organized as follows. Section II surveys the relevant research contributions. Section III describes the architecture and functionality of the JDCBL. Section IV details the results of our experimental evaluations. Finally, the conclusion is given in Section V.

II. Related Works

A. Feature Extractor

In fabric defect detection, feature extraction is crucial for establishing clear feature mapping relationships, directly impacting detection accuracy. Despite recent advancements, existing methods still face significant challenges in extracting fine-grained features, especially for elongated and subtle defects. Traditional convolutional neural networks (CNNs) such as AlexNet [21], VGG [11], and ResNet [12] excel at capturing high-level semantic features. For instance, Zhou et al. [24] proposed an automatic fabric defect detection network (Fabric Net) integrating FPNs and deformable convolutions (DCs), achieving notable results on the DAGM dataset. However, due to their fixed receptive fields, these models often struggle to capture fine details or small-sized defects, limiting their application in complex scenarios. To overcome these limitations, researchers have introduced attention-based models such as SE-Net [16], ECA-Net [17], Transformer [18], and Swin-Transformer [22]. These methods assign dynamic weights to different regions of the image, enabling the model to focus on more relevant features. For example, Zheng et al. [25] proposed SE-YOLOv5 based on YOLOv5, successfully applying SE-Net to fabric defect detection and significantly enhancing detection performance. Qu et al. [26] introduced the "U-SMR network," merging ResNet50 [12] and Swin-Transformer [22], achieving impressive results on the ZJU-Leaper dataset. However, despite improvements in feature focusing capabilities, challenges remain in simultaneously modeling local features and global context, especially when images contain highly variable or subtle defects. Recently, researchers have proposed combining convolutional networks with attention mechanisms to fully leverage the advantages of both. For instance, BoTNet [23] integrates convolution and self-attention mechanisms, significantly enhancing the model's feature extraction ability. Inspired by these methods, we propose a novel feature extraction framework that promotes self-attention modeling within images through attention mechanisms. By considering the contextual relationships of individual keys, we achieve a comprehensive representation of both static and dynamic features of the image, thereby improving the accuracy and robustness of fabric defect detection.

B. Long-Tail Learning

The long-tail learning problem in fabric defect detection can cause serious class bias during model training, which may lead to false detection and bring serious economic or safety consequences. Therefore, it is crucial to solve the long-tail

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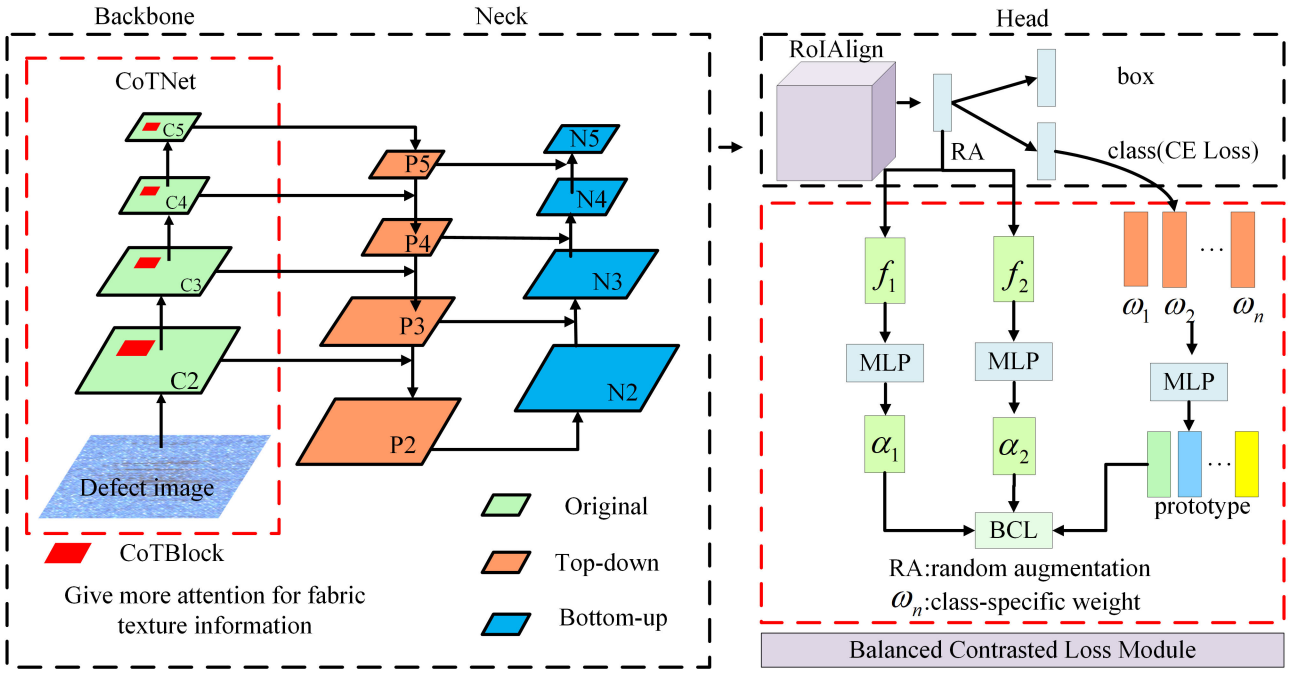


Fig. 2. Overall framework of the JDCBL consists of three main parts: backbone, neck, and head. The framework deploys CoTNet as the backbone to enhance local feature representation, generating multiscale feature maps (C2–C5). These feature maps are fed into PAFPN, which adaptively focuses on defect-related features across different scales. The head module then performs defect localization and classification using the fused features. Additionally, BCL is integrated into the training process to address class imbalance and reduce bias toward tail classes.

learning problem, especially to ensure that the model can fully learn the tail classes. At present, the main methods to solve the long-tail learning problem include class rebalancing, information augmentation, and module improvement [27]. These methods provide solutions from different perspectives to address the specific challenges in long-tail learning. The main strategy of class rebalancing is to resample the data, including oversampling and under sampling. For example, Lee et al. [28] proposed decision boundary resampling in unbalanced learning and applied it to lesion detection. Although this method improves the accuracy of tail classes, it often leads to a decrease in the accuracy of head classes and may cause overfitting problems on tail classes. Information augmentation has been a research hotspot in recent years, mainly through transfer learning and data augmentation to compensate for the problem of insufficient data. For example, Zhan et al. [2] developed a new prototype network that can achieve robust fabric defect classification under imbalanced data. Karegowda et al. [29] applied transfer learning to fabric stain detection and achieved high detection accuracy. Liu et al. [30] proposed a fabric defect detection framework based on generative adversarial networks (GANs), which effectively alleviated the data scarcity problem by generating synthetic samples. However, although the GAN-based method performed well in experiments, its high computational cost limits its widespread use in practical applications. Module improvement improves model performance through strategies such as decoupled training and classifier design. For example, Shao et al. [31] proposed a pixel-level semi-supervised fabric defect detection method based on multitask average teacher (MT), which can achieve efficient detection with a small number of labeled samples. Lu et al. [32] proposed a spatial filling strategy and applied it to fabric defect detection, achieving significant results. Although the module improvement method can effectively alleviate the class bias problem caused by data imbalance, the single method still has limitations. In addition, existing studies often focus on a few types of defect samples, while ignoring the complexity of the overall data distribution. To this end, we proposed an improved strategy that combines the advantages of multiple methods: we use prototype learning to achieve data augmentation in batch training, and combine it with a category balancing strategy to increase the exposure rate of tail category data, thereby comprehensively improving the model's ability to learn long-tail data.

III. Methodology

A. JDCBL Framework

This article explores a JDCBL framework for defect detection on fabric surfaces, and its structure illustrated in Fig. 2. This framework is based on Faster RCNN [33]. which consists of the backbone network, FPN, region proposal network (RPN), and R-CNN module. The backbone networks for feature extraction primarily include VGG [11] and ResNet [12]. However, VGG demonstrates weak spatial information processing capabilities, often resulting in the loss of critical information. Though ResNet establishes connections between initial and local features, its performance in extracting fine-grained features is unsatisfactory. These networks cannot apply effective self-attention to partial images, making them unsuitable for detecting defects in fabric with fine-grained details. CoTNet constructs self-attention mechanisms

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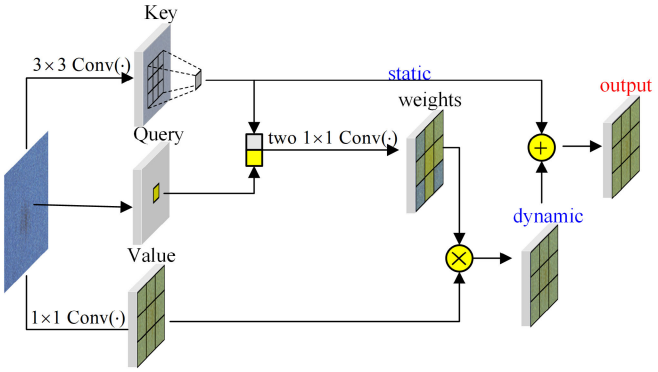


Fig. 3. Structure of CoT. The key and value of the input image are obtained through 3 × 3 convolution and 1 × 1 convolution, respectively, and then the key is concatenated with the query and the context weight of the image is obtained through two 1 × 1 convolutions. The dynamic information is multiplied by the value and fused with the key representing the static information to get the final output.

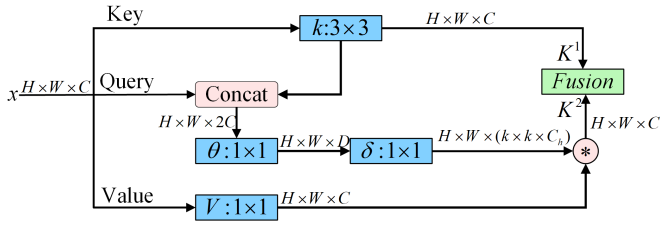


Fig. 4. Calculation details of CoT. denotes the matrix multiplication.

for images and establishes contextual relationships of local images, thus computing relational matrices between neighboring keys. It enables the fusion of static and dynamic features to develop effective local relationships and extract relevant features. These capabilities make CoTNet well-suited for fabric defect detection tasks. Additionally, to address the data imbalance issue, we integrated the BCL module into the classification component to mitigate the learning bias between head-class and tail-class data. Sections IV-B and IV-C provide a detailed explanation of CoTNet and BCL.

B. Contextual Transformer Block

The contextual transformer block (CoTBlock) enhances the representation of fine-grained features by leveraging the self-attention module from Transformers, which also encodes global information. It achieves this by utilizing the rich contextual relationships among input keys to improve self-attention learning. The structure of the CoTBlock is illustrated in Fig. 3. First, it encodes the static context of the image. The static contextual features are then concatenated with the query and processed through two 1 × 1 convolution layers to generate an attention matrix representing the relationships produce the output. This structure effectively utilizes the interaction between static and dynamic information, thus contextual information between neighboring keys, promoting enabling localized self-attention learning and representation of fine-grained features.

Fig. 4 elaborates on the working mechanism of the CoT module. The input 2-D feature $X \in R^{H\times W\times C}$ is represented as key (K), query (Q), and value (V), among them:

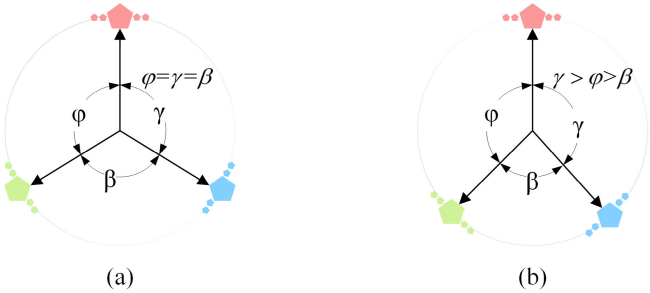


Fig. 5. Regular simplex. (a) Indicates that upon completion of the model training, all classes collapse to the vertices of a regular simplex, presenting a symmetrical configuration with equal angles. (b) Asymmetrical configuration with unequal angles.

$K = X$, $Q = X$, $V = X\mathrm{W_V}$. First, a spatial convolution operation is applied to the key in the input grid using a 3 × 3 kernel to generate static contextual features $K^1 \in R^{H\times W\times C}$. These static contextual keys K^1 are then concatenated with the Q and processed through two 1 × 1 convolution layers to compute the attention matrix

$$A = [K^1, Q]\mathrm{W}_\theta \mathrm{W}_\delta \tag{1}$$

where W_θ with ReLU [34] activation function and W_δ without activation function. We multiply the obtained attention matrix with the V, which is calculated as follows:

$$K^2 = V \otimes \mathrm{A} \tag{2}$$

where $\otimes$ denotes matrix multiplication. Finally, the static contextual information K^1 is fused with the dynamic contextual information K^2 to produce the final output enriched by contextual representation.

To validate the effectiveness of CoTBlock, we integrated it into the ResNet architecture and compared the performance of ResNet50 and CoTNet50, as shown in Table I. By replacing certain convolutional branches in ResNet with CoT blocks to enable the fusion of static and dynamic information, while keeping other parameters unchanged, experimental results demonstrated the superior feature representation capabilities of the CoTBlock.

The key to preventing inductive bias in the model lies in training an unbiased model. When training is complete and the loss function reaches its minimum, the features of various defect classes collapse to the vertices of a regular simplex, forming a symmetric configuration, as shown in Fig. 5(a). However, if the model is left unregulated during the learning process on long-tailed data, it can result in the imbalanced scenario illustrated in Fig. 5(b). Thus, it is essential to intervene in the model's learning process. First, we implement class complements to ensure effective learning for all classes within each batch. Then, we apply class average to balance the gradient contributions of head and tail classes. These two strategies are integrated into a BCL function. By optimizing the gradients of this loss function, we correct the classifier's biased learning toward different classes. The working principles of class complement and class average are detailed below.

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TABLE I
COMPARISON OF RESNET50 AND COTNET50 STRUCTURES

stage	ResNet50	CoTNet50	output
res1	$7\times7 conv, 64, stride = 2$		112×112
res2	$3\times3 maxpooling, stride = 2$		56×56
	$\begin{Bmatrix}1\times1,64\\3\times3,64\\1\times1,64\end{Bmatrix}\times3$	$\begin{Bmatrix}1\times1,64\\\text{CoT},64\\1\times1,64\end{Bmatrix}\times3$	
res3	$\begin{Bmatrix}1\times1,128\\3\times3,128\\1\times1,128\end{Bmatrix}\times4$	$\begin{Bmatrix}1\times1,128\\\text{CoT},128\\1\times1,128\end{Bmatrix}\times4$	28×28
res4	$\begin{Bmatrix}1\times1,256\\3\times3,256\\1\times1,1024\end{Bmatrix}\times6$	$\begin{Bmatrix}1\times1,256\\\text{CoT},256\\1\times1,1024\end{Bmatrix}\times6$	14×14
res5	$\begin{Bmatrix}1\times1,512\\3\times3,512\\1\times1,2048\end{Bmatrix}\times3$	$\begin{Bmatrix}1\times1,512\\\text{CoT},512\\1\times1,2048\end{Bmatrix}\times3$	7×7
	Global average pooling 20-d fc, softmax		1×1

1) Class Complement: Since the data in each mini-batch is randomly sampled, it is impossible for all classes to be present simultaneously within a single mini-batch. This results in the classifier failing to learn information about all classes during training, leading to learning bias. We define e_j as the prototype of the j class. By employing class complement, we ensure that all data classes are included in every mini-batch. This approach guarantees that each class contributes to the model's training, thereby preventing head classes from dominating the repulsion term. The loss function after applying class complement can be expressed as

$$L_c = -\frac{1}{|B_k|}\sum_{s\in\{B_k\setminus\{i\}\}\cup\{e_j\}} \log\frac{\exp(\alpha_i,\alpha_s/\tau)}{\sum_{c\in B_j\cup\{e_j\}}\exp(\alpha_i,\alpha_c)} \tag{3}$$

where B_k represents the set of samples belonging to class k, $|\cdot|$ denotes the number of samples in this set, α_i represents the feature of the i instance, α_s denotes the feature of an instance that belongs to the same class as α_i, α_c represents the set of all class instances' features, and τ represents the temperature parameter.

2) Class Average: The goal of class average is to ensure that each class's instances contribute equally in each mini-batch, so that each class has an approximately equal contribution to the optimization. In other words, class average reduces the over-contribution of head classes while emphasizing the importance of tail classes. The loss function is expressed as

$$L_a = -\frac{1}{|B_k|-1}\sum_{s\in B_k\setminus\{i\}} \log\frac{\exp(\alpha_i,\alpha_s/\tau)}{\sum_{j\in K_B}\frac{1}{|B_j|}\sum_{c\in B_k}\exp(\alpha_i,\alpha_c)} \tag{4}$$

where K_B represents the number of classes in a mini-batch B.

The final loss function obtained through class average and class complement is

$$L_{\text{ac}} = -\frac{1}{|B_k|}\sum_{s\in\{B_k\setminus\{i\}\}\cup\{e_j\}} \times\log\frac{\exp(\alpha_i,\alpha_s/\tau)}{\sum_{j\in K}\frac{1}{|B_j|+1}\sum_{c\in B_j\cup\{e_j\}}\exp(\alpha_i,\alpha_c)}. \tag{5}$$

3) Logit Adjustment: Due to data imbalance, the final output logits (log values) usually exhibit some bias. The purpose of the logit adjust function is to mitigate the bias caused by imbalanced data and correct the decision boundary during the learning process. The definition of the logit adjust function is

$$L_{\log}(k,\varphi(\alpha)) = -\lambda_k\frac{\exp(\varphi_k(\alpha)+\delta_k)}{\sum_{k'\in[K]}\exp(\varphi_{k'}(\alpha)+\delta_{k'})} \tag{6}$$

where λ_k is the factor that controls the importance of class k, and δ_k is the compensation for class k, which is related to the class frequency. We define $\lambda_k = 1$ and $\delta_k = \log \mathrm{P}_k$, and perform logit compensation during training, where P_k represents the class prior of the label. Therefore, the final loss function is

$$L_{\text{BCL}} = \kappa L_{\log} + L_{\text{ac}} \tag{7}$$

where κ is hyperparameter that control the importance and influence of $L_{\log}$.

IV. EXPERIMENTS

A. Datasets

To validate the effectiveness of our proposed method, extensive experiments were conducted on the plain fabric dataset released by the Alibaba Cloud Tianchi platform [35]. This

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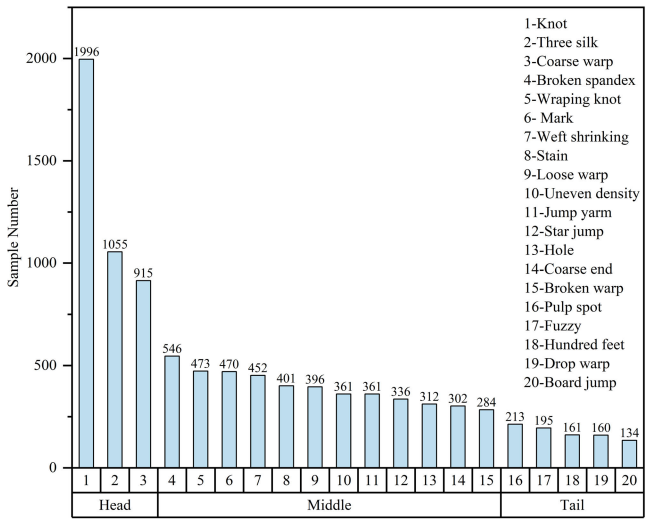


Fig. 6. Distribution of defect classes in the Tianchi fabric dataset. After processing, the dataset has 20 classes, and the data distribution is long-tailed, and the data are divided into three parts, namely head, middle, and tail to represent the head classes, slightly unbalanced, and extremely unbalanced classes.

dataset contains a total of 9523 images, including 3663 normal images and 5913 defective images, with each image having a resolution of 2446 × 1000. For this study, we only utilized the defective images in the dataset. Initially, the dataset featured 34 types of defects. To simplify the classes and focus on common features of defects, we combine defects with similar generation mechanisms and characteristics, for example, greasy stains, water stains, and stains into one class and naming stains. On the one hand, this can significantly alleviate the imbalance of data and ensure the full learning of the model. On the other hand, this can significantly improve the detection efficiency, and we end up with 20 defect types. The statistical distribution of these classes is shown in Fig. 6. Notably, the data exhibit a clear long-tailed distribution. In the diagram, we have labeled head, middle, and tail to denote head classes, slightly unbalanced classes, and severely unbalanced classes.

B. Implementations Details

The network is implemented using the PyTorch framework with Python. The experiments were conducted on an Ubuntu 22.04 GNU/Linux system, utilizing a rented Nvidia GeForce RTX 4090-D GPU. The hyperparameter settings are in Table II. The pretrained models used in our experiments are publicly available. The experiments are divided into two parts. In the first part, we compare our proposed method with the current mainstream algorithms. In the second part, we conduct ablation studies to validate the effectiveness of the improvements in the JDCBL method. Furthermore, we evaluate the performance of CoTNet and BCL within several algorithms.

C. Evaluation Metrics

To better compare the performance effects of different models, three commonly used indicators were used in the experiment: precision, recall, and variance, which were,

TABLE II
MAIN PARAMETER SETTING

Parameters	Value
Batch size	2
τ	2
Epoch	12
Initial lr	0.001
Lr	0.002
Momentum	0.9
Weight decay	0.0001
κ	0.35

respectively, defined as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

where TP, FP, and FN denote the number of true positives, false positives, and false negatives, respectively,

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}. \tag{10}$$

IoU refers to the intersection of the predicted boundary and the actual boundary divided by their union. Whether the predicted boundary box is a positive sample needs to be determined according to the predetermined threshold of the IoU. If the threshold is 0.5, all bounding boxes with IoU greater than 0.5 are positive samples, and the rest are negative samples. In this case, the AP value calculated is defined as MAP50

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} AP_i \tag{11}$$

$$\text{mAR} = \frac{1}{n}\sum_{i=1}^{n} AR_i \tag{12}$$

$$\text{mse} = \frac{1}{n}\sum_{i=1}^{n} [y_i - \hat{y}_i]^2 \tag{13}$$

where i represents the average accuracy of a particular class and n represents the quantity of the class.

D. Comparative Experiment

To evaluate the performance of JDCBL in defect detection, we compared it with other algorithms. These include RetinaNet [36], Faster RCNN [33], PAFPN [37], Mask RCNN [38], Ms RCNN [39], Libra RCNN [40], DCN [41], and Swin-Transformer [22]. In addition to Swin-Transformer, other networks use ResNet as their backbone, fixed at 50 and 101 layers, while our proposed method adopts CoTNet as the backbone. First, we quantitatively compared the model's performance on tail classes (pulp spot, fuzzy, hundred feet, drop warp, and board jump), and it is used to illustrate the performance of the model under long-tailed data, and the results shown in Table III. In the comparative experiments, our proposed method consistently outperformed others under

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TABLE III
DETECTION EFFECT OF DIFFERENT MODELS IN THE TAIL CLASS (MAP50/%)

Method	Backbone	Pulp spot	Fuzzy	Hundred feet	Drop warp	Board jump
RetinaNet [36]	ResNet50	13.0	1.9	2.3	2.7	4.8
Faster RCNN [33]	ResNet50	42.8	15.0	17.9	1.3	35.1
PAFPN [37]	ResNet50	53.1	12.5	29.6	1.4	47.3
Mask RCNN [38]	ResNet50	40.5	21.3	10.7	0.6	38.7
Libra RCNN [40]	ResNet50	36.5	10.4	0.4	0	26.4
Ms RCNN [39]	ResNet50	41.3	12.3	12.7	0.4	44.7
DCN [41]	ResNet50	58.6	15	19.6	1.5	39.9
JDCBL(Ours)	CoTNet50	**63.6**	**22.3**	**43.5**	**3.5**	**54.0**
RetinaNet [36]	ResNet101	18.1	11.1	14.9	1.9	14.5
Faster RCNN [33]	ResNet101	56.4	11.4	19.5	1.9	35.4
PAFPN [37]	ResNet101	61.0	10.7	30.7	0.8	45.0
Mask RCNN [38]	ResNet101	43.0	16.5	18.7	0.3	38.3
Libra RCNN [40]	ResNet101	37.8	11.9	0.5	0.6	35.9
Ms RCNN [39]	ResNet101	47.8	9.1	18.2	0.5	30.5
DCN [41]	ResNet101	63.1	13.6	24.6	1.0	31.0
Swin-Transformer [22]	-	63.3	18.1	25.9	1.8	43.8
JDCBL(Ours)	CoTNet101	**68.3**	**28.2**	**39.7**	**7.4**	**46.5**

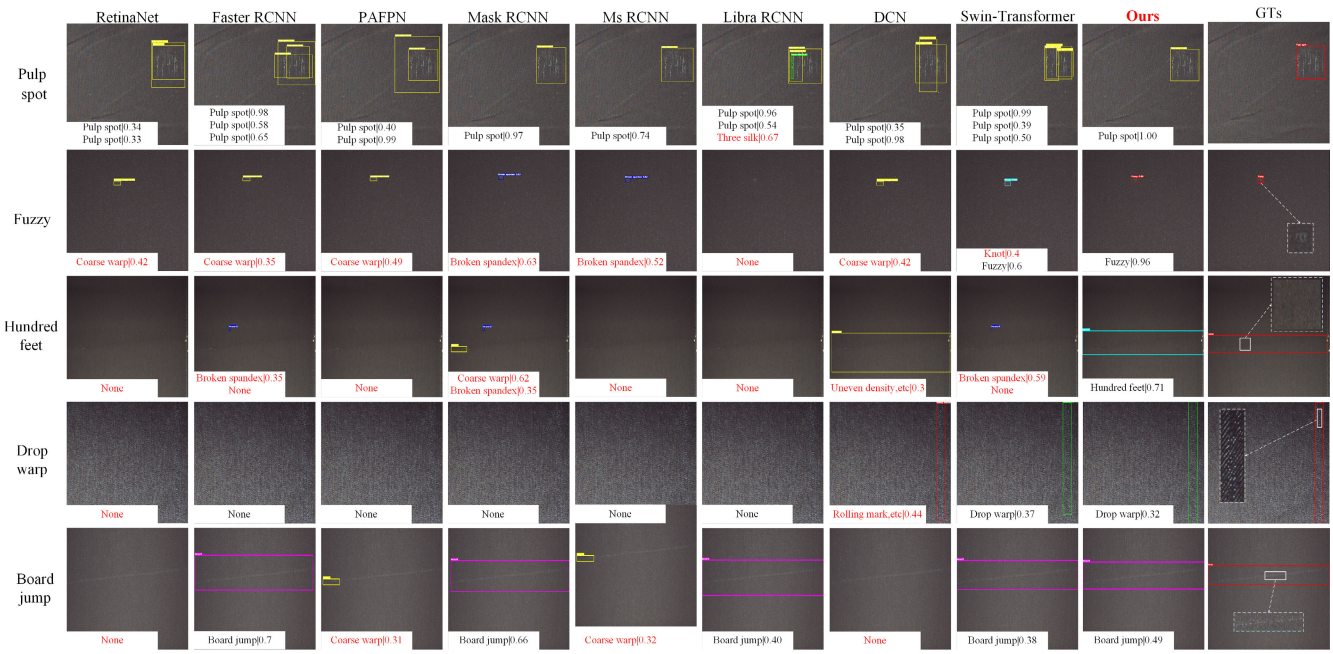


Fig. 7. Comparison of detection effects of several methods on tail classes.

the same network depth. Specifically, for fuzzy defects, our method achieved a 50 value of 28.2% with CoTNet101, over 10% higher than other methods. For 100-feet defects, the MAP50 value reached 43.5% with CoTNet50. Significant improvements were also observed for other defect classes. And then, the different model detection results are shown in Fig. 7. From the detection results, it can be observed that for Pulp spot, all methods except Mask RCNN, Ms RCNN, and our proposed approach exhibits repeated detection frames. This issue arises due to the dataset being collected in real-world industrial settings, where images contain substantial noise, causing networks to misinterpret noise as defect features. For fuzzy defects, Libra RCNN failed to detect any defects, while other comparative methods produced false positives. When detecting 100-feet and drop warp defects, multiple methods experienced missed detections, with some false positives

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TABLE IV
COMPARISON OF THE OVERALL EFFECT OF DIFFERENT MODELS

Method	Backbone	MAP (%)	MAP50 (%)	MAR (%)	MSE
RetinaNet [36]	ResNet50	7.2	18.2	16.3	3549.8
Faster RCNN [33]	ResNet50	12.0	29.6	24.4	2146.5
PAFPN [37]	ResNet50	13.5	32.1	25.9	1989.6
Mask RCNN [38]	ResNet50	12.6	29.6	26.5	2068.7
Libra RCNN [40]	ResNet50	10.6	25.5	21.9	2356.8
Ms RCNN [39]	ResNet50	12.3	28.2	25.3	2297.5
DCN [41]	ResNet50	13.3	33.6	27.3	**1572.5**
JDCBL(Ours)	CoTNet50	**15.6**	**37.2**	**28.4**	1839.9
RetinaNet [36]	ResNet101	8.8	23.5	19.7	3434.8
Faster RCNN [33]	ResNet101	13.2	32.0	25.9	1944.6
PAFPN [37]	ResNet101	14.7	33.6	27.8	1896.2
Mask RCNN [38]	ResNet101	13.0	30.7	25.9	1905.7
Libra RCNN [40]	ResNet101	11.6	28.3	21.5	2067.3
Ms RCNN [39]	ResNet101	13.2	30.5	26.9	2007.7
DCN [41]	ResNet101	13.9	33.0	26.4	**1626.7**
Swin-Transformer [22]	-	15.4	34.6	30.1	2139.9
JDCBL(Ours)	CoTNet101	**17.0**	**39.3**	**32.0**	1762.4

also observed under some methods. For board jump defects, the models demonstrated significant performance variability: Faster RCNN, Mask RCNN, and Libra RCNN could identify defects but showed large localization errors, while PAFPN and Ms RCNN could only partially localize defects and exhibited false positives, RetinaNet, and Libra RCNN do not detect any defects.

An analysis of tail-class defects reveals that these defects are often small and elongated, making them hard to detect. Although some models achieved reasonable detection performance, issues like false positives and inaccurate localization persisted. In contrast, JDCBL integrates a contextual attention mechanism into the backbone network, effectively considering the contextual relationships of pixels surrounding defects. This generates attention weights enriched with local information, enabling dynamic adjustment of pixel weights in defect regions and yielding richer defect feature representations. Furthermore, the BCL module introduces diversity in feature representation through random transformations. The class complement strategy ensures all classes are learned in each batch, while the class-balance strategy ensures balanced learning across classes. This approach increases the exposure of tail classes in extremely imbalanced datasets, thereby adjusting the model's learning weights for tail classes and improving defect detection and recognition for these classes.

It is worth noting that some long-tail learning methods improve tail-class performance at the expense of head-class performance, resulting in a decline in overall detection accuracy [27]. Therefore, to better illustrate the model's performance, we conducted a comprehensive comparison of overall performance across different models, as shown in Table IV. To demonstrate the effect of the BCL, we specifically compared the results of some long-tail methods. The results are shown in Table V. Regardless of whether the network depth was 50 or 101 layers, JDCBL consistently achieved the best results. This further confirms that JDCBL does not sacrifice detection performance for the majority of classes despite incorporating BCL. We also evaluated the model's detection results shown in Fig. 8. When detecting knot defects, Faster RCNN, PAFPN, and Ms RCNN all exhibited missed detections. For elongated and continuous defects like three silk and star jump, multiple models showed localization inaccuracies, Mask RCNN producing false positives, with RetinaNet even not detect any defect. In contrast, our method demonstrated superior performance. This indicates that the introduction of BCL did not compromise the detection accuracy for head classes. Instead, the integration of CoTNet enhanced the model's ability to identify small and elongated defects effectively. Moreover, the mse of the model shows that the robust performance of the model is well. The above comparative experiments and results conclusively demonstrate the capability and robustness of JDCBL in detecting fabric defects under conditions of data imbalance. From the results in Table V, we can see that BCL is better than other compared

TABLE V
COMPARISON OF LONG TAIL LEARNING METHODS

Method	Backbone	MAP50(%)
Focal [36]	ResNet50	23.9
SCL [42]		32.1
Logit [43]		33.5
BCL(Ours)		34.6

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TABLE VI
ABLATION ANALYSES ON FABRIC DEFECT DATASET. ✓ INDICATES ADDITION, ✗ INDICATES NO ADDITION

Depth	CoTNet	BCL		MAP50(%)
		Class-complement	Class-average	
50	✗	✗	✗	32.1
50	✓	✗	✗	33.9
50	✓	✓	✗	34.8
50	✓	✗	✓	35.5
50	✓	✓	✓	37.2

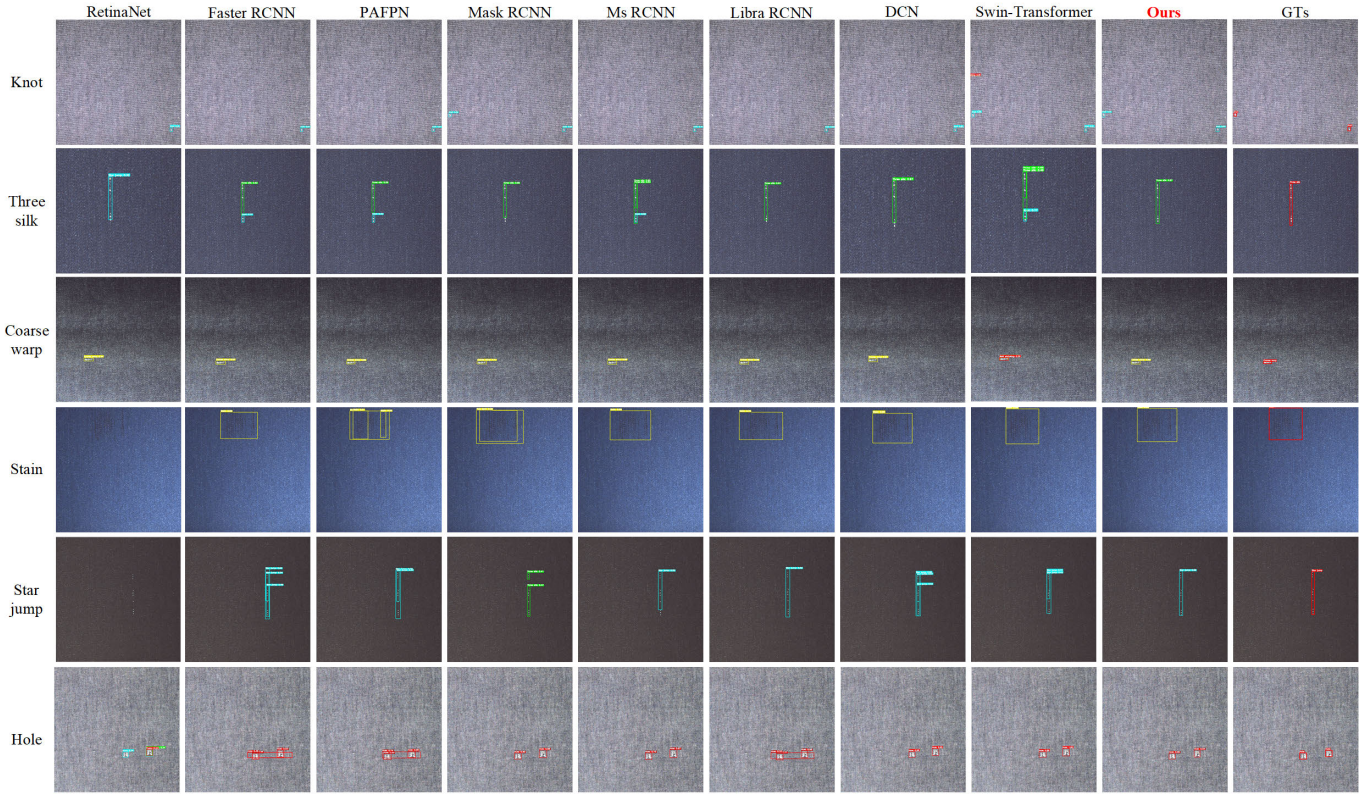


Fig. 8. Detection effects of different models on head classes and slightly imbalanced classes.

methods. Especially focal loss, which has a huge negative impact on the overall detection effect of the model. Although BCL has also been adjusted by logit, the results are significantly better than single logit after class complement and class average.

E. Ablation Experiment

To further validate the effectiveness of JDCBL in fabric defect detection, we conducted ablation experiments to evaluate the impact of each improvement on network's performance. The results are presented in Table VI. Based on the PAFPN framework with a fixed network depth of 50 layers, we incrementally introduced improvements for evaluation. The two parts of BCL, namely class complement and class average, were evaluated, respectively. First, CoTNet was introduced as the backbone network. By leveraging the contextual Transformer to enhance interactions between static and dynamic information, the representation of defect features was improved, and the overall detection performance was enhanced, demonstrating CoTNet's excellent capability in detecting small and elongated defects. Then, the class complement enabled balanced learning of all classes in each batch, increasing the influence of tail classes during training. The class average alleviated biased learning caused by data imbalance by equalizing the gradient contributions of all classes. With class average, the model could better focus on tail classes, further improving detection precision. The results showed that with CoTNet alone, the MAP50 has increased from 32.1% to 33.9%. While class complement and class average (i.e., the full BCL module) in play, the MAP50 reached 37.2%. Compared with the baseline method PAFPN, the proposed method significantly improved fabric defect detection.

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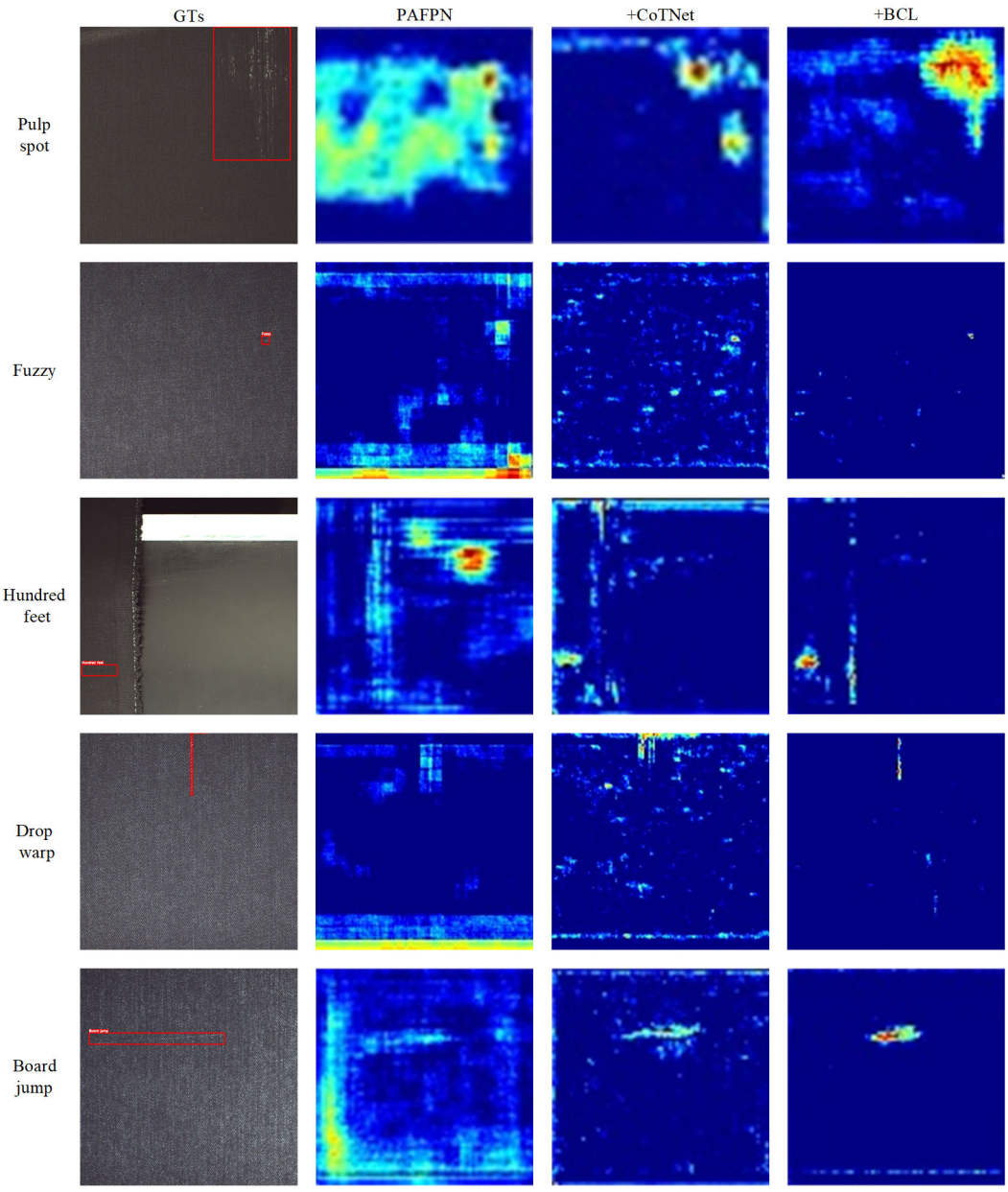


Fig. 9. Comparative heatmaps of ablation experiments.

To illustrate the impact of these improvements on feature extraction in greater detail, we used heatmaps to evaluate the defect detection results. The heatmaps were generated by fusing features across multiple scales, with different intensities indicating the network's confidence in defect regions. We selected tail-class defect images for comparison, and the heatmaps results are shown in Fig. 9. The heatmaps of the experiment demonstrate that introducing CoTNet enabled the network to focus more precisely on defect areas, thus enhancing its localization capabilities. The inclusion of BCL further adjusted the influence of tail classes, leading to more accurate localization and recognition of defects. As the experiments proceeds, the distributions on the heatmaps become increasingly concentrated on the defect areas. This improvement was evident for both small and elongated defects, highlighting the enhanced learning ability of the model for defect features. However, certain improvements may exhibit better adaptability to specific models. Therefore, we evaluated the adaptability of CoTNet and BCL across various two-stage algorithms. By independently incorporating these improvements, we assessed their impact on model performance. The results are summarized in Tables VII and VIII. As shown in Tables VII and VIII, CoTNet and BCL consistently improved detection accuracy across different models. For instance, when CoTNet was added independently, the MAP50 of PAFPN increased from 32.1% to 33.9%, while Libra RCNN saw a 1.8% improvement. When BCL was added alone, the MAP50 of Libra RCNN increased from 26.5% to 30.5%, achieving a 4% improvement. These results demonstrate the excellent adaptability of both improvements, which not only excel in

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TABLE VII
EVALUATE OF COTNET (MAP50/%)

Method	Original	+CoTNet
RetinaNet [36]	18.2	21.3
Faster RCNN [33]	29.6	31.0
PAFPN [37]	32.1	33.9
Mask RCNN [38]	29.6	29.8
Libra RCNN [40]	26.5	28.3
MS RCNN [39]	28.2	29.7
DCN [41]	33.6	34.9

TABLE VIII
EVALUATE OF BCL (MAP50/%)

Method	Original	+BCL
RetinaNet [36]	18.2	20.1
Faster RCNN [33]	29.6	31.1
PAFPN [37]	32.1	34.1
Mask RCNN [38]	29.6	31.4
Libra RCNN [40]	26.5	30.5
MS RCNN [39]	28.2	28.6
DCN [41]	33.6	34.7
Swin-Transformer [22]	34.6	35.4

JDCBL but also deliver remarkable performance gains in other networks.

V. CONCLUSION

This article proposes a new JDCBL framework for fabric defect detection in real scenarios. The framework uses the CoTNet as its backbone network to extract fine-grained features. Within CoTNet, the self-attention mechanisms enabled by the Transformer technology encodes global information and establish connections between keys through context, thus facilitating effective interaction between static and dynamic information. Additionally, BCL is designed as a loss function to correct model learning preferences and increase exposure of tail classes by employing class completion and class average, thereby achieving balanced learning between head and tail classes. The effectiveness of CoTNet and BCL was evaluated on a fabric dataset released on the Tianchi platform, and the ablation experiment confirmed these two improvements' effectiveness. The experimental results show that the proposed JDCBL framework surpasses compared methods in defect localization and classification, achieving MAP50 of 39.3%, which is more than 5% higher than other comparative methods.

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